Phi Lam

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Education

University of British Columbia Vancouver, BC
Bachelor of Applied Science - Electrical Engineering *Graduation May 2025*
Relevant Coursework: Data Structures, Algorithms, System Software, Microcomputers, Networking, Deep Learning, Applied ML

Work Experience

Tesla Motors Sparks, NV
Software Engineering Intern *Jun. 2022 - Dec. 2022*

- Designed and deployed system software in **Go** for production testers, implementing state machine logic, I/O drivers, and fault injection to support validation and mass deployment of **100K+** Powerwall 3 units in year one
- Collaborated with Gigafactory Shanghai test team to deliver start-of-production ready software for two Powerwall 3 production testers, completing validation, release notes, and handover
- Designed and deployed backend software updates using REST APIs and Grafana to deliver real-time health metrics for **100K+** deployed devices and improve system reliability
- Built data pipelines to query and process cell-level battery metrics, eliminating redundant test stages and saving **$200K+** in downstream processing costs

Firmware/ Embedded Software Engineering Intern *Feb. 2022 - Jun. 2022*

- Drove firmware architecture for an R&D battery module emulator, collaborating with internal business units to resolve critical design flaws and project savings of **$3M**
- Developed a backend communication pipeline between user applications and embedded devices using Go, Protobuf, and MQTT, enabling precise hardware control through a software interface
- Designed and tested a fault injection framework to simulate hardware-level failures in firmware, enabling validation of recovery logic and improving system resilience during edge-case scenarios

General Motors Remote
Software Integration Engineering Intern *Jan. 2024 - Aug. 2024*

- Analyzed **1,000+** vehicle data logs using Vspy to identify preliminary root causes of software issues across critical vehicle programs, reducing diagnostics-related product launch delays by ∼**10%**
- Built internal data tool to classify and categorize diagnostic test code issues, accelerating workflows by ∼**5%**

Extracurriculars and Projects

UBC Formula Electric Vancouver, BC
Lead Software Engineer, Vehicle Controls *Jan. 2023 - May 2025*

- Drove development of vehicle dynamics control algorithms in **C/C++** for a newly designed 4-WD electric race car in a large multi-contributor codebase while preserving backward compatibility across firwmare layers
- Owned end-to-end development of regenerative braking logic in **C/C++** with real-time embedded constraints
- Improved battery efficiency by ∼**10%** through regenerative braking tuning and improved average lap time by ∼**12%**
- Led torque vectoring design strategies in **Matlab** and **Python**, while supervising a subteam of 3 engineers focused on design, testing, integration, and optimization

GPU Computing Software Stack
Software Engineer *Sept. 2023 - May 2024*

- Built a **distributed** job-scheduling system on **GCP** with a central control plane and **Dockerized** worker nodes
- Wrote a **Bash/systemd** agent on user devices that detects new jobs in shared **NFS** folders and posts them to a **Flask** backend control plane for scheduling, resource balancing, and monitoring
- Built **REST APIs** and a **React** dashboard for real-time job tracking and logs, backed by **MariaDB** and **PostgreSQL (TimescaleDB)** for users, state management, and run metadata

Technical Skills

Languages: Python, C/C++, Go, JS, Verilog, SQL, NoSQL
Software Tools/Frameworks: Linux/Bash, Git, React, AWS/GCP, Docker, MATLAB Simulink, Wireshark